Università degli Studi di Padova
Quantum Information with Atoms and Photons
Master degree of Physics of Data

Exam assignment

Cavity blockade: a way to make photons interact

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Contents

1 Physical Meaning of $\hat{H}_d$ **2**

2 Hamiltonian in the Drive Frame and RWA **2**

3 Numerical Solution: No Coupling ($\tilde{g} = 0$) **2**

4 Steady State with Coupling ($\tilde{g} \neq 0$) **2**

5 Occupation of Two-Photon States **2**

6 Two-Photon Correlation Function $g^{(2)}$ **2**

7 Frequency Scan and Graphical Representation **2**

8 Bonus: Analytical Predictions **2**

General Topic

In this exercise you will explore a regime in which a cavity, once filled with a single photon, blocks the possibility to insert another one. This can be effectively interpreted as a photon-photon interaction mediated by the atom in the cavity, a possibility that photons in vacuum do not have (the photon-photon cross section predicted by QED in vacuum is extremely small). The first observation of this phenomenon was done in Birnbaum et al. Nature 436, 87 (2005).

A driven cavity is described by a Jaynes-Cummings model:

$$\hat{H} = \hbar\omega_c \hat{a}^\dagger \hat{a} + \hbar\omega_a |e\rangle\langle e| + \hbar\tilde{g}(\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger) + \hat{H}_d,$$

where $\hat{H}_d = Ee^{i\omega t}\hat{a} + \text{h.c.}$ is a (pumping) driving field.

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